

Online 1

Question:

1. Using only MOV, ADD, SUB, INC, DEC, and NEG, translate the following high-level language assignment statements into assembly language.

(7)

$$B = 3A - B + 2C$$

Take A as user input, also show a message

Declare B as byte variable and initialize it with 3

Declare C as constant and initialize it with 1

The output should be printed using an appropriate message.

Assume the final result is a single digit.

Sample Input:	Sample Output:
Enter the value of A: 2	B = 5

2. Write a program to take a lowercase letter as input and convert it to uppercase letter with appropriate message (3)

Sample Input:	Sample Output:
Enter a lowercase letter: a	The uppercase letter: A

Online 2

Question:

1. Write an assembly program to print the following pattern in the console.

Sample Input	Sample Output
5	54321 5432 543 54 5
7	7654321 765432 76543 7654 765 76 7

2. Write an assembly code to do the following with the value of BH register:
 - If the third bit (from the right) in the **BH register** is **0**, apply a mask to the value stored in the BH register in such a way that **the whole number gets complimented. Then count the number of 1's in the lower nibble of the masked value.**
 - If the third bit (from the right) in the BH register is **1**, count the **number of 0's in the upper nibble** of the BH register. Then **multiply the total number of zeroes by 5 and store the product in a variable name Result. (Do not use the MUL keyword.)**

Note: **Only display the count of 1 and 0 in the console. You can only use add, logical, shift and rotate instructions to solve this.**

Sample value	Sample final value	Explanation
BH = 193d	Result : Not applicable here	BH = 11000001b (3rd bit from right = 0) Masked value = 00111110b No. of 1's in lower nibble of masked value = 3
BH = 207d	Result = 0AH	BH = 11001111b (3rd bit from right = 1) No. of 0's in the upper nibble = 2 Result = 2 * 5 = 10d = 0AH

Online 3

Write an 8086 Assembly program to take two numbers, X and Y, as input and find their GCD using the following algorithm:

1. Divide X by Y.
2. If the remainder is 0, stop. Y is the GCD of X and Y.
3. Otherwise, replace X by Y and Y by the remainder.
4. Repeat from Step 1.

Display the GCD on a new line along with appropriate messages.

Sample Input	Sample Output	Explanation
X: 48 Y: 18	GCD: 6	48 / 18 = 2, Remainder = 12 18 / 12 = 1, Remainder = 6 12 / 6 = 2, Remainder = 0 So, GCD = 6
X: 56 Y: 42	GCD: 14	56 / 42 = 1, Remainder = 14 42 / 14 = 3, Remainder = 0 So, GCD = 14